# DEPARTMENT OF COMPUTER SCIENCE, IT & ANIMATION

## A PROJECT SYNOPSIS

## On

### Social Media

## ContentXplorer Dashboard


**Guided by**

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## Project Allotment Letter

This is to declare that the Project entitled **"Social Media ContentXplorer Dashboard"** has been allotted to **'Prerna.D.Pohakar', 'Pooja.V.Palodkar'** in partial fulfillment of the degree of **BCA(Science)** of Dr. B. A. M. University during the academic year 2025-2026 .

| **Project Guide** | **In-charge** | **HOD** |
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# INDEX

# INTRODUCTION

**ContentXplorer** is a Power BI–based analytics dashboard designed to solve this challenge by providing a unified view of social media content performance across multiple platforms. The dashboard analyzes key metrics such as views, likes, comments, shares, and engagement rate to identify trending and high-performing content. By transforming raw social media data into clear and interactive visual insights, ContentXplorer helps creators and organizations make data-driven decisions to improve content strategy and audience engagement.

- Social media platforms produce a large volume of content every day, making performance analysis complex and time-consuming.
- Content creators and organizations often struggle to identify which posts are trending and which platform performs best.
- ContentXplorer is a Power BI–based dashboard designed to analyze and compare social media content across multiple platforms.
- The dashboard uses key metrics such as views, likes, comments, shares, and engagement rate for evaluation.
- It visually highlights top-performing and trending content using interactive charts and KPIs.
- The project transforms raw social media data into meaningful insights that support data-driven content decisions

# OBJECTIVE

The primary objective of the ContentXplorer project is to design and develop an interactive Power BI dashboard that enables comprehensive analysis of social media content performance across multiple platforms. The dashboard aims to identify trending, popular, and high-performing content by visualizing engagement metrics and providing actionable insights for data-driven decision-making.

**1. To Centralize Social Media Content Data**

- To collect and organize social media content data from multiple platforms such as Instagram, YouTube, Facebook, TikTok, LinkedIn, and X (Twitter).
- To create a unified dataset that allows easy comparison of content performance across different platforms.
- To eliminate the complexity of analyzing data separately for each social media platform.

**2. To Analyze Content Performance Metrics**

- To analyze key engagement metrics such as views, likes, comments, and shares.
- To calculate important performance indicators such as engagement rate and trending score using Power BI DAX formulas.
- To evaluate how different metrics contribute to overall content popularity.

**3. To Identify Trending and High-Performing Content**

- To identify top-performing posts based on engagement and trending scores.
- To rank content using customized performance measures instead of relying on single metrics like likes or views.
- To highlight viral and trending content for quick decision-making.

**4. To Compare Social Media Platforms**

- To compare engagement levels across multiple social media platforms.
- To determine which platform generates the highest reach and audience interaction.

- To understand platform-specific content behavior and performance trends.

**5. To Analyze Content Types and Categories**

- To evaluate performance based on content type such as videos, reels, shorts, posts, and images.
- To analyze category-wise performance including education, entertainment, travel, food, fashion, and technology.
- To identify which type of content works best for specific platforms.

**6. To Study Audience Engagement Patterns**

- To analyze engagement behavior across different posting times and days.
- To identify the best time and best day to publish content for maximum reach.
- To observe audience interaction trends using time-based analysis.

**7. To Provide Interactive Data Visualization**

- To design visually appealing and interactive dashboards using Power BI.
- To use charts, graphs, KPI cards, slicers, and drill-through features for deeper analysis.
- To allow users to filter and explore data dynamically.

**8. To Enable Data-Driven Decision Making**

- To help content creators and marketers make informed decisions based on analytics rather than assumptions.
- To support content planning strategies by identifying successful patterns from historical data.
- To reduce content failure by learning from past performance insights.

**9. To Improve Content Strategy and Optimization**

- To identify weak-performing content and suggest improvement opportunities.
- To analyze which engagement factors influence content success the most.

- To assist users in optimizing future content strategies for better engagement.

**10. To Demonstrate Practical Use of Power BI**

- To showcase the effective use of Power BI features such as Power Query, DAX measures, and data modeling.
- To demonstrate real-world application of business intelligence tools.
- To strengthen analytical and visualization skills through hands-on dashboard development.

**11. To Support Scalability and Future Enhancements**

- To design the dashboard in a way that supports future data growth.
- To allow integration with additional social media platforms in the future.
- To prepare the system for advanced analytics such as Python or machine learning integration if required.

**12. To Provide Academic and Professional Value**

- To fulfill academic requirements for data analytics and business intelligence projects.
- To create a project that reflects real-world industry use cases.
- To develop a portfolio-ready dashboard suitable for interviews and professional presentations.

# Expected Contribution & Future Scope

The **ContentXplorer – Social Media Analytics Dashboard** developed using **Power BI** provides significant value by converting large volumes of social media data into clear, visual, and actionable insights. The project supports efficient analysis and informed decision-making for content creators, marketing teams, and organizations.

**1.Centralized Performance Analysis**

- The dashboard brings social media content data from multiple platforms into a single Power BI environment.
- This centralized approach simplifies analysis and eliminates the need to review each platform separately.

**2. Actionable Insights for Decision-Making**

- Power BI visualizations such as KPI cards, charts, and tables help users quickly identify high-performing and under-performing content.
- The dashboard enables informed decisions regarding content planning and optimization.

**3. Improved Content Strategy Development**

- By analyzing metrics such as views, likes, comments, shares, and engagement rate, the project helps identify successful content patterns.
- Users can adjust their content strategies based on historical performance insights.

**4. Cross-Platform Content Comparison**

- The dashboard allows direct comparison of content performance across different social media platforms.
- This helps organizations determine which platform provides higher reach and engagement.

**5. Enhanced Data Visualization and Reporting**

- Interactive Power BI visuals enable dynamic filtering and drill-down analysis.
- Users can explore data from summary-level insights to post-level details.

## Future Scope of the Project

### 1. Real-Time Data Refresh

- Future versions can include scheduled or automatic data refresh.
- This will ensure updated insights as new content data becomes available.

### 2. Advanced DAX Calculations

- Additional DAX measures can be created to improve performance evaluation.
- More refined metrics such as weighted engagement scores and growth indicators can be introduced.

### 3. Enhanced Drill-Through and Tooltips

- Advanced drill-through pages can provide deeper insights at the individual post level.
- Custom tooltips can improve user understanding of metrics.

### 4. Integration with Additional Data Sources

- The dashboard can be extended to include more social media platforms or external datasets.
- This will increase the analytical scope of the project.

### 5. Improved Visual Design and User Experience

- Future improvements can focus on layout optimization, color themes, and usability.
- Enhanced visuals will make insights easier to interpret.

# SCOPE OF PROJECT

The scope of the ContentXplorer – Social Media Analytics Dashboard defines the boundaries, features, and limitations of the project. The project is developed exclusively using Power BI and focuses on analyzing historical social media content data to generate meaningful insights.

## Functional Scope

### 1 Social Media Content Analysis

- The project analyzes social media content data from multiple platforms such as Instagram, YouTube, Facebook, TikTok, LinkedIn, and X (Twitter).
- It evaluates content performance based on key metrics including views, likes, comments, shares, and engagement rate.

### 2 Cross-Platform Performance Comparison

- The dashboard enables comparison of content performance across different social media platforms.
- Users can identify which platform provides higher engagement and reach.

### 3 Trending Content Identification

- The project identifies trending and high-performing content using calculated measures such as engagement rate and trending score.
- Ranking and sorting features help highlight top-performing posts.

## Technical Scope

### 1 Power BI Development

- The entire project is developed using Power BI for data modeling, visualization, and reporting.
- Power Query is used for data cleaning and transformation**.**

### 2 DAX Calculations

- DAX formulas are used to calculate key performance indicators such as engagement rate, trending score, and platform rankings.

- These calculations enhance analytical depth within the dashboard.

**3 Data Visualization**

- The project uses various Power BI visuals such as KPI cards, bar charts, line charts, tables, and heat maps.
- Visual design focuses on clarity, readability, and ease of interpretation**.**

## User Scope

**1 Target Users**

- Content creators
- Social media managers
- Digital marketing teams
- Students and data analysts

**3.2 User Benefits**

- Quick access to performance insights
- Simplified content analysis
- Improved decision-making
- Reduced manual reporting effort

# PROJECT REQUIREMENT SPECIFICATION

## a) Hardware Requirement

To develop and run the ContentXplorer – Social Media Performance Dashboard, the following hardware is required:

- **Laptop Model:** HP Victus
- **Processor (CPU):**
  AMD Ryzen 5 / Intel Core i5 (or higher)
- **RAM:**
  16 GB RAM (sufficient for Power BI and data analysis tasks)
- **Storage:**
  512 GB SSD (ensures fast data loading and smooth performance)
- **Graphics:**
  Dedicated Graphics (NVIDIA GTX / RTX series) or Integrated Graphics
- **Display:**
  15.6-inch Full HD Display (1920 × 1080 resolution)
- **Input Devices:**
  Keyboard and Mouse / Touchpad
- **Internet Connection:**
  Required for downloading tools, datasets, and sharing reports

## b)Software Requirement

The following software tools are required for the successful development of the project:

- **Operating System:**
  Windows 10 / Windows 11 (64-bit)
- **Data Visualisation Tool:**
  Microsoft Power BI Desktop
- **Data Source Tools:**
  Microsoft Excel / CSV files
- **Supporting Software:**
  1. Web Browser (Chrome / Edge)
  2. Microsoft Office (optional, for documentation)

# METHODOLOGY

ContentXplorer – Social Media Analytics Dashboard using Power BI. The project follows a structured process that includes data collection, data cleaning and transformation, data modeling, and visualization design to ensure accurate and meaningful insights.

## a)Data Collection:

The data for the ContentXplorer – Social Media Performance Dashboard was collected from the Kaggle website, which provides publicly available datasets for data analysis and learning purposes.
The dataset included social media performance metrics such as platform name, content type, views, likes, shares, comments, engagement rate, and date.

After downloading the dataset from Kaggle, the data was imported into SQL Server using SQL Server Management Studio (SSMS).
The data was stored in relational tables and basic validation was performed to ensure data accuracy.

Power BI Desktop was then connected to SQL Server as the data source to fetch the data for analysis and visualization.
Using SQL Server as an intermediate database helped in better data management, structured storage, and efficient querying.

## b)Data Cleaning & Transformation (ETL): Mention the tools/techniques used (e.g., Power Query).

The ETL (Extract, Transform, Load) process was carried out to prepare the data for analysis.

- **Extract:**
  Data was extracted from SQL Server, where the dataset imported from Kaggle was stored using SQL Server Management Studio (SSMS).
- **Transform:**
  Data cleaning and transformation were performed using Power Query Editor in Power BI.
  The following techniques were applied:
  - Removal of duplicate records
  - Handling missing and null values
  - Standardizing data formats (dates, text, numeric values)

  - Renaming columns for better readability
  - Creating calculated columns where required
  - Filtering irrelevant or unused data

- **Load:**
  The cleaned and transformed data was loaded into Power BI for data modeling and visualization.

## c)Data Modeling:

Data modeling was performed to structure the data efficiently and enable accurate analysis.

- Relationships were created between tables based on common fields such as Post ID, Platform, and Date.
- Measures and calculated columns were created using DAX (Data Analysis Expressions) to compute key performance indicators.
- Proper data relationships ensured correct aggregation and filtering across visuals.

## d)Visualization & Dashboard Design:

**Types of Visuals Used**

- KPI Cards to display key metrics such as total views, likes, shares, comments, and engagement rate
- Bar Charts to compare performance across different social media platforms
- Column Charts to analyze content type performance (videos, reels, posts, shorts)
- Line Charts to show engagement and performance trends over time
- Pie / Donut Charts to represent content distribution across platforms
- Tables to show detailed performance data
- Slicers for filtering data by platform, content type, and time period

**Design Principles Applied**

- Used a clean and minimal layout to avoid visual clutter
- Maintained consistent color themes and fonts across all visuals
- Focused on key KPIs at the top for quick insights
- Ensured proper alignment and spacing for better readability
- Applied interactive filtering so users can easily explore data
- Designed visuals to support storytelling and trend analysis

# INTRODUCTION TO POWER BI DESKTOP, POWER BI SERVICE

## Introduction to Power BI Desktop

Power BI Desktop is a Windows-based business intelligence tool used to connect, clean, analyze, and visualize data.It allows users to create interactive reports and dashboards using data from multiple sources such as Excel, SQL Server, and online services.

**In this project, Power BI Desktop was used for:**

- Connecting to SQL Server as a data source
- Performing data cleaning and transformation using Power Query
- Creating data models and DAX measures
- Designing interactive reports and dashboards

Power BI Desktop is mainly used for development and report creation.

## Introduction to Power BI Service

Power BI Service is a cloud-based platform provided by Microsoft that is used to publish, share, and collaborate on Power BI reports.

**In this project, Power BI Service can be used for:**

- Publishing reports created in Power BI Desktop
- Sharing dashboards with users through web access
- Accessing reports from any device with internet connectivity
- Managing datasets and refresh schedules

Power BI Service is mainly used for report sharing and collaboration.

# DASHBOARDS

A dashboard is a visual display that presents key information and performance indicators in a single view. It helps users quickly understand data, identify trends, and make informed decisions.

In the ContentXplorer project, the dashboard was created using Power BI to provide a consolidated view of social media performance across multiple platforms.

**Dashboard Page 1: Overview (KPI Dashboard)**

The first page serves as the overall summary dashboard of the project.
It displays all the key performance indicators (KPIs) such as total views, total likes, total comments, total shares, engagement rate, and total content count across all social media platforms.

This page also includes interactive buttons and navigation options that allow users to move easily between different social media platform dashboards such as YouTube, Instagram, and Facebook. The purpose of this page is to provide a quick snapshot of overall performance and help users understand the general trend of content performance across platforms.

**Dashboard Page 2: YouTube Platform Analysis**

The second page focuses specifically on YouTube performance analysis.
It includes platform-specific visuals such as bar charts, line charts, tables, and KPIs related to YouTube metrics like views, likes, comments, subscribers, and content performance.

Slicers are provided for filtering data based on content type, time period, or channel-related attributes. Each visual is clearly labeled to help users understand which YouTube content is performing better and which trends are emerging on this platform.

**Dashboard Page 3: Instagram Platform Analysis**

The third page is dedicated to Instagram performance analysis.
This page follows a structure similar to the YouTube dashboard, ensuring consistency across platforms. It displays metrics such as likes, comments, reach, impressions, engagement rate, and content performance.

Interactive slicers allow users to filter data based on post type, time period, or campaign. This page helps users identify trending Instagram content and analyze audience engagement patterns.

**Dashboard Page 4: Facebook Platform Analysis**

The fourth page presents Facebook platform insights.
It includes visuals representing likes, shares, comments, reach, engagement, and content performance. Like other platform-specific pages, this dashboard also contains slicers and KPIs for better data exploration.

The uniform design across YouTube, Instagram, and Facebook dashboards helps users easily compare individual platform performance while maintaining a smooth user experience.

**Dashboard Page 5: Cross-Platform Comparison**

The final page focuses on the comparison of all social media platforms.
It combines data from YouTube, Instagram, and Facebook to compare overall performance metrics such as total engagement, views, content reach, and engagement rate.

Comparative visuals like clustered bar charts, tables, and KPIs are used to highlight which platform performs best for different metrics. This page helps in identifying the most effective social media platform and understanding where content strategies perform better.

# TABLES

## 1. Platform Table

- Fields / Attributes:
  - Platform_ID (PK) → Unique ID for each social media platform
  - Platform → Name of the social media platform (e.g., Instagram, YouTube, Facebook)

## 2. Content Table

- Fields / Attributes:
  - PostID (PK) → Unique ID for each post/content
  - PostType → Type of content (Video, Reel, Post, Short)
  - Category → Category of content (e.g., Entertainment, Education)
  - PostDate → Date when content was posted
  - Platform_ID (FK) → Links content to a specific platform

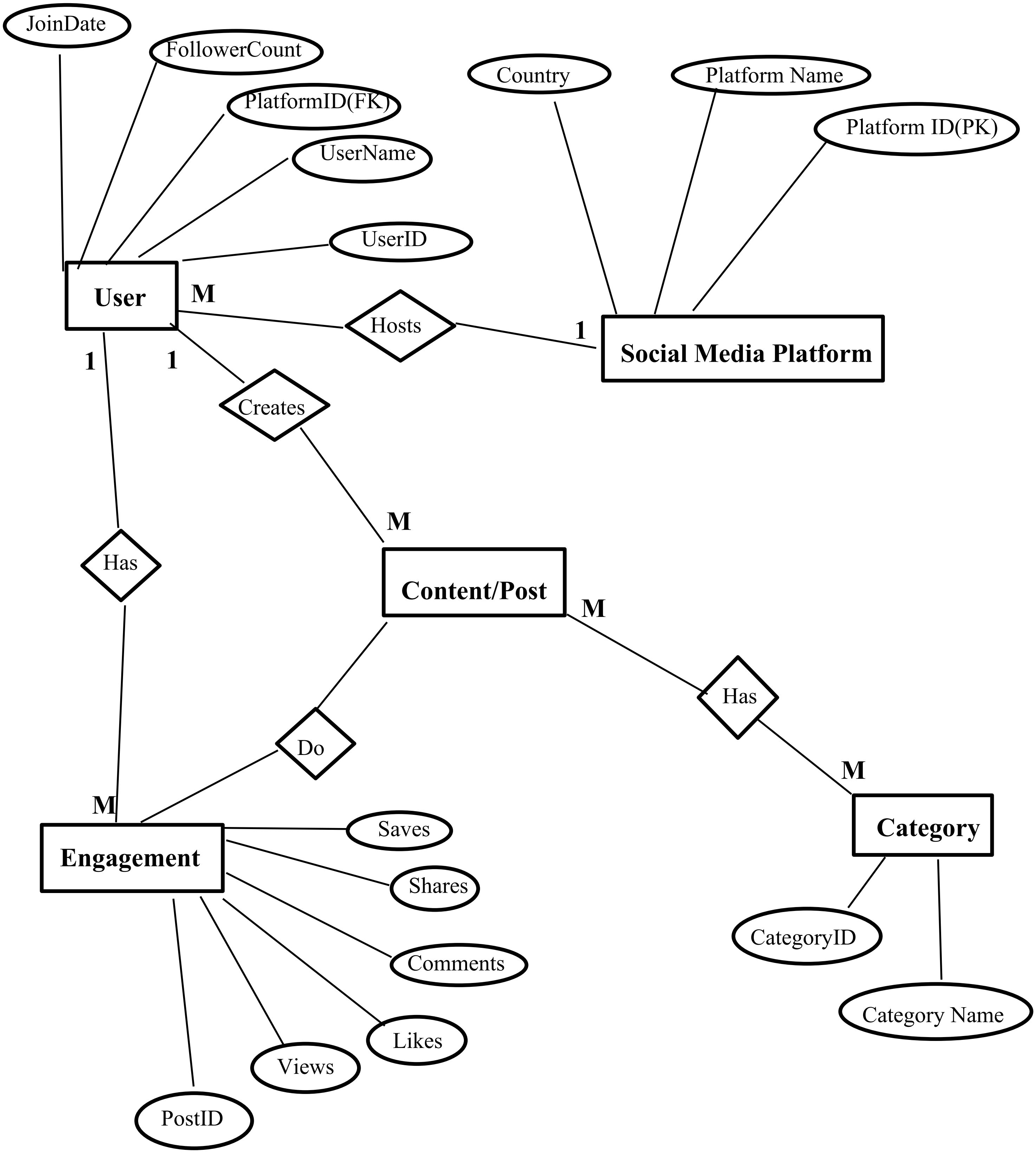
## 3. Performance Table

- Fields / Attributes:
  - Performance_ID (PK) → Unique ID for each performance record
  - PostID (FK) → Links performance metrics to specific content
  - FollowerCount → Followers of the platform at that time
  - Views → Number of times the post was viewed
  - Impressions → Total impressions of the post
  - Reach → Total unique users who saw the post
  - Likes, Comments, Shares, Saves → Engagement metrics
  - ProfileVisits → Number of profile visits from that post
  - WatchTimeSeconds → Time users spent watching the content

- BounceRate → Percentage of users who left quickly
- Engagement → Combined engagement score
- CTR (Click-Through Rate) → Percentage of clicks from impressions
- TrendingScore → Score indicating how trending the post is

# DIAGRAMS

## 1. E-R Diagram



## 2. Data Flow Diagram